ViKAG-Faith: Graph-Grounded Claim-Level Faithfulness Evaluation and Verification for Vietnamese Knowledge-Augmented Generation

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***Abstract*—Vietnamese RAG and GraphRAG question answering systems are usually scored with F1 or a single LLM-judge score for the whole answer. Neither shows which statements are backed by evidence and which are invented or contradicted. For legal regulations, one wrong effective date or fine is enough to cause harm. We build ViKAG-Faith, a claim-level faithfulness evaluation pipeline on ViHERMES. Two systems, Naive RAG and GraphRAG, answer 100 questions of 1–5 hops with Qwen2.5-3B. Answers are split into 610 claims, of which 284 claims from 80 answers are labelled supported, contradicted or unsupported against a common evidence set. We compare RAGAS, LLM-judges, SelfCheckGPT, multilingual NLI verifiers with and without graph information, number and date rules, verifiers fine-tuned on synthetic in-domain data, and a verify-then-revise loop. The primary labels are silver labels assigned by an LLM (Claude); 165 claims were also labelled blindly by one human annotator. On those claims, the 7B judge reaches the highest macro-F1 (0.528), above every NLI verifier (best 0.463). Fine-tuning with synthetic data raises the mDeBERTa verifier from 0.404 to 0.521 on silver labels, yet the gain disappears on human labels (0.381 versus 0.409). Silver labels agree with the human on 65% of claims ($\kappa = 0.29$), and metric rankings change with the label source. Verify-then-revise halves the contradiction rate but lowers token F1, because the small generator mostly deletes flagged claims. Training on hard negatives shaped like the observed errors gives the best silver score (0.643) but no gain on human labels, and a two-tier checker matches the judge on human labels while sending it only half of the claims. A fourth round adds a 32B judge, which is closer to the LLM labels but further from the human, three more systems on which the 7B judge ranks systems almost opposite to the silver labels, a 30-question pilot in a second domain, and a KAG-style Verify operator that reduces wrong claims mainly by refusing to answer.**

***Index Terms*—GraphRAG, faithfulness, claim-level verification, Vietnamese, ViHERMES, meta-evaluation**

I. Introduction

GraphRAG and knowledge-augmented generation (KAG) bring knowledge graphs into retrieval to answer multi-step questions [1]–[4]. ViHERMES [5] provides a multi-hop benchmark for Vietnamese healthcare regulations together with a legal structure graph, and shows that removing its Auditor (verification) module lowers the LLM-judge score from 0.7554 to 0.6823. Verification clearly matters, but the way faithfulness itself is measured on Vietnamese has not been evaluated.

An F1 or a single judge score for a whole answer does not reveal which statement is wrong. Common faithfulness metrics such as RAGAS [6], LLM-as-a-judge [7] and SelfCheckGPT [8] have been validated mainly on English. DSC2025 ViHallu [9] evaluates Vietnamese hallucination, but not in connection with a knowledge graph or retrieved evidence.

Our approach splits answers into atomic claims, as FActScore [10] does, checks each claim against a common evidence set of retrieved passages, gold passages and graph information, and measures how well each automatic metric matches reference labels. This is a meta-evaluation: the object under test is the checker, not the chatbot. We ask three questions.

- **RQ1:** How well do existing faithfulness metrics (RAGAS, LLM-judge, SelfCheckGPT, zero-shot NLI) match reference labels on Vietnamese?
- **RQ2:** Do graph information, number and date rules, and fine-tuning on synthetic in-domain data help a verifier detect wrong claims better than a generic NLI verifier?
- **RQ3:** How much does a verify-then-revise loop reduce unsupported claims, and does it hurt answer quality?

Our contributions are: (C1) a reproducible claim-level evaluation pipeline on ViHERMES; (C2) the pilot ViFaithKAG dataset with 284 silver-labelled claims, 165 human-labelled claims and the scores of every metric; (C3) a meta-evaluation of 24 metric variants, 12 in each of two rounds, with ablations of graph input, rules, synthetic data and revision rounds; and (C4) an analysis of LLM label reliability against a human annotator, showing that metric rankings depend on the label source.

II. Related Work

Measuring RAG faithfulness. RAGAS [6] defines faithfulness as the share of statements that can be inferred from the context, using an LLM to extract and judge statements. FActScore [10] scores each atomic fact against a knowledge source. LLM-as-a-judge [7] uses a strong LLM as the grader; agreement with humans depends on model size and domain.

Hallucination detection. SelfCheckGPT [8] needs no evidence and relies on consistency between sampled answers. SemEval-2024 SHROOM [11] and SemEval-2025 Mu-SHROOM [12] standardise the detection of invented spans, and DSC2025 ViHallu [9] is a Vietnamese challenge on this topic.

Vietnamese and multilingual NLI. XNLI [13] includes Vietnamese, and ViNLI [14] is a Vietnamese NLI corpus that is part of VLUE [15]. mDeBERTa-v3 [16], trained on multilingual NLI [17], is a widely used encoder verifier.

Graph RAG, KAG and regulations. RAG [18], GraphRAG [2], LightRAG [3], HippoRAG 2 [4] and KAG [1] use graphs for retrieval or reasoning. ViHERMES [5] builds a Document–Article–Clause graph with amendment, replacement and citation relations for healthcare documents.

Table I compares these lines of work with ours.

Positioning. We do not propose a new QA system. We evaluate verification metrics on Vietnamese at claim level against human labels, keeping the ViHERMES data and graph unchanged and adding only an evaluation layer.

III. Proposed Method

Fig. 1 summarises the pipeline. The upper row generates answers with two systems, the lower row evaluates each claim, and the revise branch uses verification results to rewrite answers.

A. Question answering systems

Both systems share the ViHERMES corpus (998 passages from 471 documents), a hybrid BM25 [19] and bge-m3 [20] retriever (weight 0.5), a budget of 6,000 characters and the Qwen2.5-3B-Instruct generator [21] at temperature 0. Naive RAG takes the 8 best passages. GraphRAG takes 4 seed passages, propagates over the ViHERMES graph (depth up to 4, at most 8 nodes) and adds a list of document relations to the prompt. Any difference between the systems therefore comes only from how context is obtained.

B. Claim extraction and common evidence

Qwen2.5-3B extracts claims with a Vietnamese prompt containing two examples. Each claim must have one main predicate and stand alone, with pronouns replaced by names and numbers, dates and document identifiers kept intact. For each answer, the evidence is the union of the retrieved passages and the gold passages of the question, with the graph information of each passage (document title, identifier and relations). Every metric and the annotator use exactly this set, so differences between metrics reflect metric quality and not retrieval quality.

C. Labels and answer-level metrics

Each claim receives one label from supported, contradicted and unsupported. For an answer with claim set C, the faithfulness score GGF is the share of supported claims, and CR and UR are the contradicted and unsupported rates:

$$\mathrm{GGF} = \frac{|\{c \in C : c \text{ supported}\}|}{|C|}, \tag{1}$$

$$\mathrm{CR} = \frac{|\{c \in C : c \text{ contradicted}\}|}{|C|}, \tag{2}$$

$$\mathrm{UR} = \frac{|\{c \in C : c \text{ unsupported}\}|}{|C|}. \tag{3}$$

D. Checkers

LLM-based metrics. RAGAS faithfulness is reimplemented from the two original prompts of the library, and a claim-level LLM-judge assigns the three labels to our claims. Both run with Qwen2.5-3B and 7B through Ollama; round 2 adds 14B, round 4 adds 32B, and a Qwen 3B verifier that returns a label with an evidence span. Unreadable outputs are excluded from measurement and never defaulted to a label.

NLI verifier. mDeBERTa-v3-base-xnli-multilingual [16], [17] runs on windows of 900 characters with a stride of 700; round 2 adds XLM-R large xnli, zero-shot and fine-tuned on 16,000 pairs. For claim c and windows w,

$$P_{\sup}(c) = \max_{w} P(\text{entail} \mid w, c), \tag{4}$$

$$P_{\mathrm{con}}(c) = \max_{w \in W_2(c)} P(\text{contradiction} \mid w, c), \tag{5}$$

where $W_2(c)$ holds the two windows with the highest word overlap with the claim, because NLI models tend to return contradiction for unrelated text. A claim is supported if $P_{\sup} \geq \tau_s$, otherwise contradicted if $P_{\mathrm{con}} \geq \tau_c$, and otherwise unsupported. The thresholds are chosen on the dev split (V1 zero-shot: $\tau_s = 0.65$, $\tau_c = 0.45$). V1 uses passages only. V2 adds the document title, identifier and relations to the start of every window. V3 adds at most 3 triples sharing keywords with the claim, from 467 graph triples, and, through a mutual index, the source passage of any document the claim cites.

Fine-tuning on synthetic data. We combine 12,000 ViNLI [14] pairs with 3,328 pairs built from passages outside the evaluation sample: supported is the original sentence; contradicted changes a number or date or adds a negation; unsupported is a sentence from another passage of the same document. Training runs one epoch at learning rate 5×10^{-6}, batch size 4 with 4 accumulation steps, in float32, keeping the best dev checkpoint. An ablation uses ViNLI only. Round 2 repeats this with 12,000 ViNLI pairs and 17,993 synthetic pairs. A first run on a 4 GB GPU had to shorten training premises to 450 characters and 200 tokens; it was then repeated with the round 1 input of 900 characters and 320 tokens.

Rules and SelfCheckGPT. A number, date or identifier in a claim that is missing from the evidence demotes supported to unsupported; if the matching evidence sentence holds a different value of the same kind, the claim becomes contradicted. SelfCheckGPT-NLI [8] samples 3 extra answers at temperature 0.7 from the same prompt and context, and averages the mDeBERTa entailment and contradiction probabilities using each sample as premise. It does not see the evidence.

E. Verify-then-revise

For RQ3, the 7B judge flags unsupported or contradicted claims. Qwen2.5-3B receives the same retrieved context, the old answer and the flagged claims, and rewrites the answer by fixing or dropping them. The new answer is split into claims and rescored by the 7B judge, for at most 2 rounds.

TABLE I
RELATED WORK COMPARED BY EVALUATION UNIT, EVIDENCE, LANGUAGE AND USE OF A GRAPH

Work	Unit	Evidence	Language	Graph
RAGAS [6]	Statement	Retrieved context	Mainly English	No
FActScore [10]	Atomic fact	Knowledge source	English	No
SelfCheckGPT [8]	Sentence	None, sampled answers	English	No
SHROOM, Mu-SHROOM [11], [12]	Span	None required	English, multilingual	No
ViHallu [9]	–	Not tied to retrieval	Vietnamese	No
ViHERMES [5]	Whole answer	Retrieved context	Vietnamese	Legal graph
This work	Claim	Common evidence set	Vietnamese	Legal graph, triples

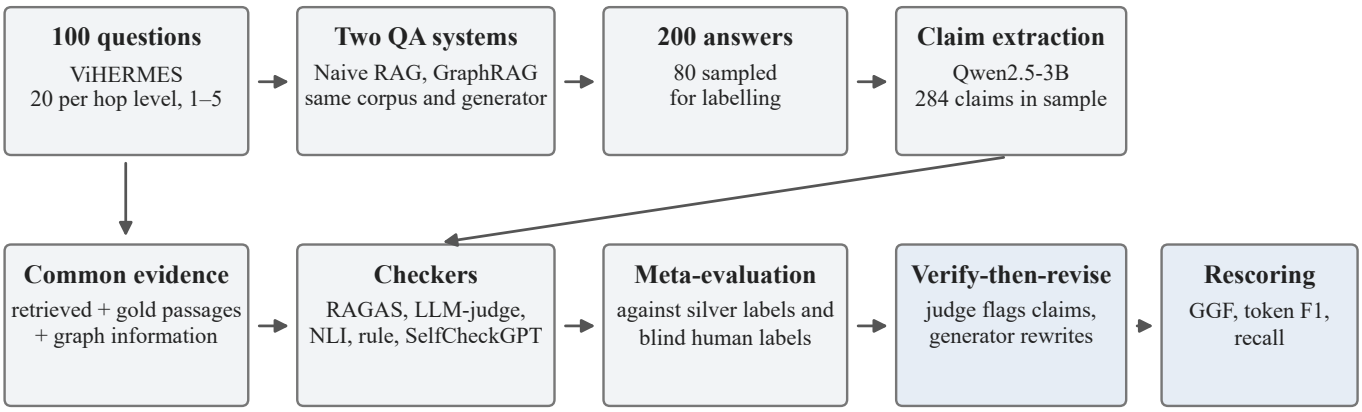


Fig. 1. The ViKAG-Faith pipeline. Answers in the labelled sample go to claim extraction, then to the checkers and to meta-evaluation.

TABLE II
DATASET STATISTICS

Item	Value
Questions / hop levels	100 / 20 per level, 1–5
Answers (two systems)	200
Labelled sample	80 answers, 284 claims (15 malformed claims excluded)
Silver label distribution	supported 184, contradicted 42, unsupported 43
Dev / test split by question	51 / 218 claims

TABLE III
VERIFIER ACCURACY ON THE DEV SET (THREE LABELS)

Verifier	ViNLI dev	Synthetic dev
mDeBERTa zero-shot	0.758	0.545
Fine-tuned, ViNLI + synthetic	0.793	0.875
Fine-tuned, ViNLI only	0.798	0.588

IV. DATASET

We draw 100 of the 1,561 ViHERMES questions by stratified sampling (20 per hop level, seed 42), so two systems give 200 answers. The labelled sample has 40 questions (8 per hop level) for 80 answers and 284 claims (Table II).

Claim extraction quality. In the labelled sample, an answer has 3.55 claims on average and at most 10, with little change across hop levels (3.06 to 3.88). 15 of 284 claims (5.3%) were excluded as malformed: repeated questions, lead-in phrases without content, vague statements, or placeholders such as "[?]". A further 12 claims hold three or more numeric values and were marked for the annotator, because such claims can be partly right. Extraction by the same small model thus adds a source of error before any checker runs.

Synthetic pairs come from passages outside the evaluation sample, so they do not leak into the test split. The dev set holds 1,000 ViNLI pairs and 10% of the synthetic pairs. Table III shows that synthetic data strongly raises accuracy on synthetic pairs, while the ViNLI-only model barely learns the in-domain error types.

Silver labels. Claude labelled all claims with evidence passage numbers and reasons. These labels are the reference for every test-set measurement, but they are not human labels.

Human labels (blind audit). One annotator worked on an offline HTML page without seeing Claude's labels or reasons, and without knowing which system produced an answer. Pass C covers 60 claims stratified by Claude's label (30/15/15) with full passages. Passes G, H, I and J each cover 30 random claims not used before, one claim per screen with the 4 most overlapping evidence sentences and the option to open the full passage. A pre-filled review pass (A) and two passes with exports not created through the labelling page (D, E) were excluded (Table IV).

Table V shows three claims where the human label equals Claude's label, translated from Vietnamese.

TABLE IV
HUMAN LABELLING PASSES

Pass	Selection and display	Claims	Agree	κ
C	Stratified, full passage	60	60%	0.29
G, H, I, J	Random, 4 sentences	105	69%	0.30
All		165	65%	0.29
Excluded	A, D, E	–	–	–

TABLE V
EXAMPLE CLAIMS AND EVIDENCE (TRANSLATED)

Claim	Evidence	Label
The total fine of the three clinics is 8 million VND.	One clinic alone was fined 30,000,000 VND.	contra.
The Population Department has no role in inspection or handling violations.	Its duties include specialised inspection and handling violations in the population field.	contra.
Circular 30/2017/TT-BYT takes effect on 01/01/2018.	The evidence gives no such date.	unsup.

V. EXPERIMENTAL SETUP

Configurations. Round 1 compares 12 variants: LLM-judge and RAGAS with 3B and 7B; NLI zero-shot, fine-tuned and ViNLI-only, each with V1 and V2; SelfCheckGPT-NLI; and rule only. Each NLI variant is also tested with and without rules, and an always-supported baseline is included. Round 2 adds 12 variants: judge and RAGAS 14B, zero-shot V3, fine-tuned V1 and V3 at 450 and 900 characters, XLM-R zero-shot and fine-tuned V1 and V3, and Qwen 3B with spans. Round 3 adds XLM-R trained with error-shaped hard negatives and a two-tier checker. Round 4 adds judge and RAGAS 32B, three more systems, a pilot in a second domain and a KAG-style Verify operator.

Metrics. At claim level we report three-label macro-F1, per-label F1 and AUROC for detecting non-supported claims. At answer level we report Spearman and Kendall correlation between automatic scores and reference GGF on 61–63 test answers, depending on valid outputs. For revision we report changes in GGF, CR, UR, token F1, exact match and token recall against ViHERMES answers.

Statistics. We use 95% bootstrap confidence intervals with 1,000 resamples, paired bootstrap for macro-F1 differences, paired Wilcoxon tests for system and before/after comparisons, and Cohen's κ for agreement. Intervals are not corrected for multiple comparisons.

Hardware. Models run on a Google Colab T4 GPU (Ollama with Q4_K_M quantisation; PyTorch in float32), except zero-shot V3 on CPU and the short-premise fine-tune on a local GTX 1650. The 32B judge and the round 4 systems run on Kaggle with two T4 GPUs.

VI. RESULTS

Table VI and Fig. 2 give the main results of round 1.

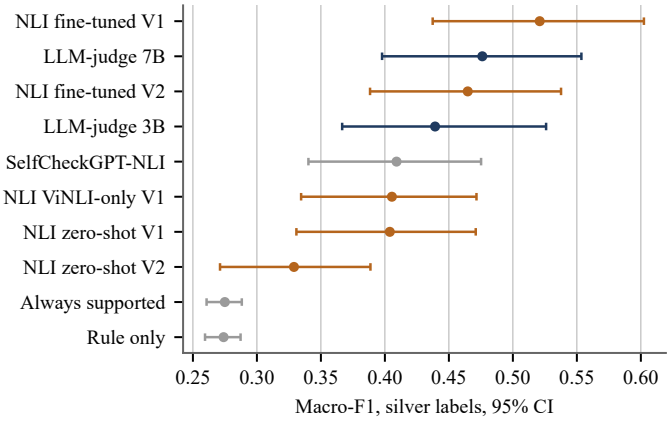


Fig. 2. Claim-level macro-F1 on the test split. Blue: LLM-judges; orange: NLI verifiers; grey: SelfCheckGPT, rule and baseline.

A. RQ1: Existing metrics

In round 1, among methods that need no training, the 7B judge is best at claim level (macro-F1 0.476, AUROC 0.748), ahead of the 3B judge (0.439) and zero-shot NLI (0.404). It catches 41/64 wrong claims and raises 22/153 false alarms. All methods are weak on the two error labels (Table VII). SelfCheckGPT-NLI reaches 0.409, level with zero-shot NLI without seeing the evidence, but its accuracy is low (0.463) because it treats correct claims as doubtful when samples are worded differently. At answer level, RAGAS 7B has the highest correlation with reference GGF ($\rho = 0.484$) while RAGAS 3B reaches only 0.258, so judge size matters more than the metric formula. High accuracy does not imply a good metric: always predicting supported gives accuracy 0.702, which is why macro-F1 is our main measure.

B. RQ2: Graph, rules and fine-tuning

Adding graph information to every window (V2) lowers zero-shot macro-F1 from 0.404 to 0.329. Its best threshold is very low ($\tau_s = 0.15$): with a document title at the start of each window, the model gives high entailment to any claim naming that document. Adding only triples related to each claim (V3) raises zero-shot macro-F1 to 0.423, above V2, but on human labels V3 falls below V1 (0.328 versus 0.409). Rules barely change results (0.404 to 0.401), since many errors concern the wrong subject and not a number. On silver labels, fine-tuned V1 beats zero-shot (difference 0.117, 95% CI [0.03, 0.20]) and is not significantly above the 7B judge. The ViNLI-only ablation shows no gain (0.002), so the improvement comes from the synthetic in-domain data (Table VIII).

Human labels. Fig. 3 reverses the conclusion on fine-tuning. On 165 claims, fine-tuned V1 reaches 0.381, level with zero-shot (0.409) and clearly below the 7B judge (0.528). The rank correlation of metrics between the two label sets is only $\rho = 0.46$. Verifier thresholds are chosen on Claude's labels, so one explanation is that the fine-tuned model learnt Claude's strict labelling; Section VI-F tests this. Table IX supports this: Claude labels 59 claims contradicted or unsupported, and the

TABLE VI
MAIN RESULTS OF ROUND 1. MACRO-F1, CI AND AUROC ON 218 TEST CLAIMS (SILVER LABELS); HUMAN COLUMN ON 165 HUMAN-LABELLED CLAIMS; SPEARMAN ρ BETWEEN ANSWER SCORES AND REFERENCE GGF

Method	Macro-F1	95% CI	AUROC	Macro-F1 (human)	Spearman ρ
LLM-judge Qwen2.5-7B	0.476	[0.40, 0.55]	0.748	0.528	0.418
LLM-judge Qwen2.5-3B	0.439	[0.37, 0.53]	0.643	0.349	0.367
NLI fine-tuned, V1	0.521	[0.44, 0.60]	0.738	0.381	0.142
NLI fine-tuned, V2 (+graph)	0.465	[0.39, 0.54]	0.735	0.336	0.383
NLI fine-tuned ViNLI only, V1	0.405	[0.33, 0.47]	0.649	0.395	−0.005
NLI zero-shot, V1	0.404	[0.33, 0.47]	0.545	0.409	0.003
NLI zero-shot V1 + rule	0.401	[0.33, 0.47]	0.534	0.413	0.012
NLI zero-shot, V2 (+graph)	0.329	[0.27, 0.39]	0.534	0.295	0.161
SelfCheckGPT-NLI	0.409	[0.34, 0.48]	0.600	0.336	0.270
Rule only	0.274	[0.26, 0.29]	0.513	0.310	0.147
Always supported	0.275	[0.26, 0.29]	0.500	0.279	–
RAGAS (Qwen2.5-7B)	–	–	–	–	0.484
RAGAS (Qwen2.5-3B)	–	–	–	–	0.258

TABLE VII
F1 PER LABEL ON THE TEST SPLIT (SILVER LABELS)

Method	Supp.	Contra.	Unsup.	Acc.
Judge 7B	0.853	0.321	0.254	0.687
Judge 3B	0.825	0.129	0.364	0.692
Fine-tuned V1	0.773	0.379	0.410	0.656
Zero-shot V1	0.737	0.308	0.167	0.592
SelfCheckGPT	0.565	0.400	0.262	0.463
Rule only	0.822	0.000	0.000	0.688

TABLE VIII
PAIRED COMPARISON OF MACRO-F1 ON THE TEST SPLIT (SILVER LABELS, 1,000 BOOTSTRAP RESAMPLES)

Comparison	Diff.	95% CI	Share ≤ 0
Fine-tuned V1 − judge 7B	0.047	[−0.06, 0.16]	0.188
Fine-tuned V1 − zero-shot V1	0.117	[0.03, 0.20]	0.006
Fine-tuned V1 − ViNLI only	0.115	[0.05, 0.18]	0.000
ViNLI only − zero-shot V1	0.002	[−0.06, 0.06]	0.475
Fine-tuned V1 − fine-tuned V2	0.056	[−0.01, 0.12]	0.056
Judge 7B − SelfCheckGPT	0.065	[−0.05, 0.17]	0.120

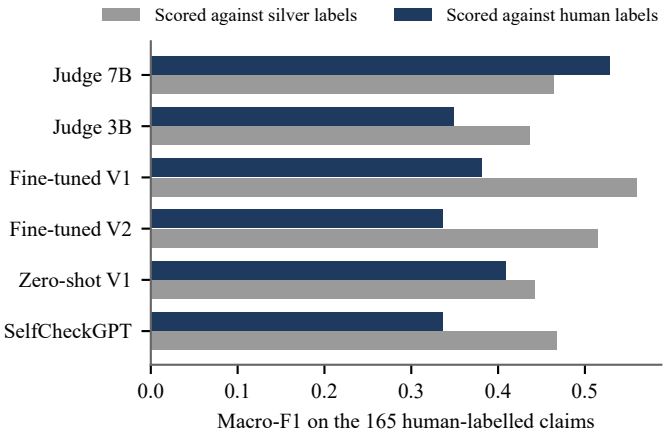


Fig. 3. Macro-F1 of metrics on the human-labelled claims, scored against each label set.

TABLE IX
CONFUSION BETWEEN CLAUDE LABELS (ROWS) AND HUMAN LABELS (COLUMNS) ON 165 CLAIMS

Claude	Supp.	Contra.	Unsup.	Agree
supported	90	5	11	85%
contradicted	16	8	6	27%
unsupported	13	6	10	34%

human agrees on only 18 and labels 29 of them supported, whereas 85% of Claude's supported labels are confirmed.

Round 2: larger models. The 14B judge reaches 0.541 on silver labels, not significantly different from the 7B judge (0.069, CI [−0.05, 0.18]). On human labels it is lower (0.439 versus 0.528, CI of the difference [−0.20, 0.03]). At answer level, RAGAS 14B reaches $\rho = 0.644$ and the 14B judge 0.597, above the 7B versions (0.484 and 0.418), but on 61 answers and without a test of the difference. XLM-R zero-shot V1 reaches 0.532 on silver labels, clearly above mDeBERTa zero-shot (0.128, CI [0.03, 0.22]), but only 0.354 on human labels. XLM-R fine-tuned V1 reaches 0.463 on human labels, significantly above XLM-R zero-shot (0.109, CI [0.02, 0.20]) and not significantly different from the 7B judge (CI [−0.17, 0.04]), at the cost of 64/153 false alarms. XLM-R fine-tuned V3 has the highest AUROC (0.821), above round 1 on silver labels (CI [0.01, 0.16]) but not on human labels. Qwen 3B with spans reaches 0.361 on silver and 0.399 on human labels, clearly below the 7B judge; only 130/263 labels carry a span copied from the evidence, ignoring case and punctuation.

Round 2: about 20,000 synthetic pairs. Accuracy on synthetic dev pairs rises from 0.594 to 0.897. With short premises (450 characters, 200 tokens), fine-tuned V1 reaches only 0.441, below round 1 (−0.080, CI [−0.16, 0.00]). Retraining with the round 1 input (900 characters, 320 tokens) gives 0.572, the highest silver score, significantly above the short-premise model (0.131, CI [0.05, 0.22]) but not above round 1 (0.051, CI [−0.02, 0.12]). AUROC barely moves (0.738 to 0.754, CI [−0.01, 0.04]). The first drop thus came from short premises, and more data did not improve discrimination. On human labels this model reaches only 0.356, clearly below the

TABLE X
ACCURACY ON 200 HELD-OUT HARD PAIRS PER ERROR TYPE

Error type	XLM-R zero-shot	XLM-R + hard negatives
Wrong agency	59%	100%
Inserted list item	60%	98%
False denial	100%	100%

7B judge (CI [−0.28, −0.05]). Table XII lists these results.

Round 3: error-shaped hard negatives. The error analysis showed that real wrong claims name the wrong agency, insert list items or deny what the text says, while our synthetic pairs only changed numbers, dates or negation. We therefore generated 3,546 pairs of these three kinds from passages outside the evaluation sample, each with a supported counterpart (an agency swapped from a fixed list of 22 health agencies, a list item inserted or removed, and a false or true statement that the passage mentions an identifier, a date, an amount or an agency), and retrained XLM-R on 16,000 pairs of which 2,946 are hard; the other 600 hard pairs are held out. Table X shows that the model learns these error types: on held-out hard pairs, accuracy on wrong agencies rises from 59% to 100%. False denials were already solved by the zero-shot model, so that template is too easy to teach anything. On real claims (Table XI), the new model reaches 0.643 on silver labels, the highest score in this study and significantly above XLM-R of round 2 (0.144, CI [0.09, 0.20]) and above the 7B judge (0.165, CI [0.06, 0.27]). On human labels it reaches 0.450, no better than round 2 (−0.013, CI [−0.09, 0.07]), and AUROC does not change significantly (CI [0.00, 0.05]). This is the third time the same pattern appears: each training change that raises the silver score leaves the human-label score where it was.

C. RQ3: Verify-then-revise

Of 80 answers, 42 have flagged claims (Table XIII). GGF rises from 0.755 to 0.830 (Wilcoxon $p = 0.014$) and CR halves, but token F1 falls from 0.595 to 0.563 ($p = 0.001$), claims per answer fall from 3.55 to 2.44, and 7 answers become "no information found". The first revision round gives most of the gain: on the 27 answers that needed a second round, GGF moves from 0.491 to 0.540 without significance ($p = 0.312$). Exact match is 0 in every configuration because ViHERMES answers are long (about 115 words), so token recall measures content loss. The GGF gain is an upper bound, since the 7B judge both flags and rescores.

Table XIV splits the loop by system. Both systems gain GGF and lose token F1, so the trade-off does not depend on the retrieval method.

Table XV gives one 3-hop example. The 7B judge flagged the claim that the amended Pharmacy Law would go to the National Assembly in 10/2025, while the document says 10/2022. The new answer reached GGF of 1 by the judge but dropped all three dates, including the correct 01/07/2025 date that both the judge and Claude labelled supported, and replaced them with a generic sentence saying the dates are stated in the context. This pattern explains how GGF rises while token recall falls.

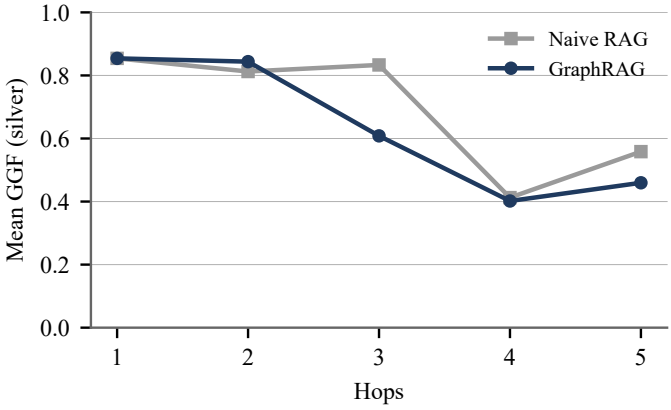


Fig. 4. Mean GGF by hop count, with 8 questions per level. GraphRAG has 7 answers with claims at 5 hops.

D. Systems and hop difficulty

The two systems do not differ significantly (paired Wilcoxon on GGF, $p = 0.191$; Table XVI). The released ViHERMES graph is coarse, with few amendment edges, so propagation adds few gold passages. Faithfulness falls clearly at 4 and 5 hops (Fig. 4).

E. Error analysis

- **Wrong dates:** an answer places the amended Pharmacy Law in 10/2025, while the document says 10/2022; 10/2025 belongs to the Food Safety Law.
- **Wrong subject:** a task of the Ministry Office is assigned to the Institute of Strategy and Policy. Number and date rules cannot catch this type.
- **Wrong aggregation:** fines from several decisions are added incorrectly. Both LLM-judges miss this because they must add numbers across documents; NLI catches it.
- **Invented lists:** an answer lists Instagram, TikTok and YouTube although the advertising certificate names others; zero-shot NLI is misled because another document in the context mentions Instagram.
- **False refusals and extraction errors:** answers claim there is no information when the evidence is explicit, and 15 claims were malformed by extraction.

Table XVII shows four of these errors with the verdict of each checker. The fine-tuned verifier matches the silver label on all four, the 7B judge on two and the 3B judge on one. The 7B judge misses the wrong sum and flags the invented item under another label.

F. Post hoc analyses

The analyses in this subsection were designed after the main results, on the same data, so they suggest explanations and do not confirm them.

TABLE XI
ROUND 3 ON THE TEST SPLIT: XLM-R TRAINED WITH ERROR-SHAPED HARD NEGATIVES, COMPARED WITH XLM-R OF ROUND 2 AND THE 7B JUDGE

Method	Silver	95% CI	AUROC	Caught	False alarms	Human
Judge 7B	0.476	[0.40, 0.55]	0.748	41/64	22/153	0.528
XLM-R fine-tuned V1, round 2	0.500	[0.42, 0.57]	0.798	58/65	64/153	0.463
XLM-R fine-tuned V3, round 2	0.477	[0.40, 0.55]	0.821	60/65	62/153	0.433
XLM-R + hard negatives, V1	0.643	[0.56, 0.71]	0.823	55/65	43/153	0.450
XLM-R + hard negatives, V3	0.626	[0.54, 0.70]	0.837	58/65	47/153	0.451

TABLE XII
ROUND 2 ON THE TEST SPLIT COMPARED WITH ROUND 1 AND THE 7B JUDGE. THRESHOLDS ARE CHOSEN ON DEV WITH THE ROUND 1 GRID. "CAUGHT" COUNTS WRONG CLAIMS DETECTED; "FALSE ALARMS" COUNTS CORRECT CLAIMS FLAGGED

Method	Silver	95% CI	AUROC	Caught	False alarms	Human
Judge 7B	0.476	[0.40, 0.55]	0.748	41/64	22/153	0.528
Judge 14B	0.541	[0.46, 0.62]	0.761	45/65	26/153	0.439
Fine-tuned V1, round 1	0.521	[0.44, 0.60]	0.738	34/65	37/153	0.381
Fine-tuned V1, 450 chars	0.441	[0.36, 0.53]	0.753	16/65	15/153	0.371
Fine-tuned V3, 450 chars	0.501	[0.41, 0.58]	0.745	32/65	31/153	0.404
Fine-tuned V1, 900 chars	0.572	[0.48, 0.64]	0.754	44/65	48/153	0.356
Fine-tuned V3, 900 chars	0.559	[0.48, 0.63]	0.734	50/65	57/153	0.346
Zero-shot V3	0.423	[0.35, 0.49]	0.558	28/65	51/153	0.328
XLM-R zero-shot V1	0.532	[0.45, 0.61]	0.743	33/65	36/153	0.354
XLM-R zero-shot V3	0.516	[0.42, 0.59]	0.758	34/65	38/153	0.370
XLM-R fine-tuned V1	0.500	[0.42, 0.57]	0.798	58/65	64/153	0.463
XLM-R fine-tuned V3	0.477	[0.40, 0.55]	0.821	60/65	62/153	0.433
Qwen 3B with spans	0.361	[0.30, 0.43]	0.601	23/65	23/151	0.399

TABLE XIII
VERIFY-THEN-REVISE ON 80 ANSWERS. GGF, CR AND UR FROM THE 7B JUDGE; F1, RECALL AND EM AGAINST VIHERMES ANSWERS

Setting	GGF	CR	UR	F1	Recall	EM	Claims
No revision	0.755	0.088	0.157	0.595	0.591	0.000	3.55
At most 1 round	0.801	0.059	0.140	0.569	0.545	0.000	2.75
At most 2 rounds	0.830	0.041	0.129	0.563	0.535	0.000	2.44

TABLE XIV
VERIFY-THEN-REVISE BY SYSTEM, AT MOST 2 ROUNDS

System	Revised	GGF before	After	F1 before	After	Refused
Naive RAG	22/40	0.707	0.810	0.601	0.552	4
GraphRAG	20/40	0.802	0.849	0.590	0.574	3

TABLE XV
VERIFY-THEN-REVISE EXAMPLE (3 HOPS, NAIVE RAG, TRANSLATED)

Before revision	After one round
The amended Pharmacy Law is expected to go to the National Assembly in 10/2025. Decree 124/2021/ND-CP on administrative penalties in health was issued in 12/2021. Autonomous hospitals may decide drug procurement under Law 90/2025/QH15 from 01/07/2025.	No information was found on when the amended Pharmacy Law is expected to go to the National Assembly. The issuing date of Decree 124/2021/ND-CP and the date autonomous hospitals may decide drug procurement under Law 90/2025/QH15 are stated clearly in the context.

TABLE XVI
ANSWER QUALITY AND FAITHFULNESS OF THE TWO SYSTEMS

System	Token F1	Gold recall	GGF	CR	UR
Naive RAG	0.570	47.0%	0.694	0.182	0.124
GraphRAG	0.563	47.9%	0.638	0.199	0.163

Difficulty by hop. Table XVIII shows that the share of non-supported silver labels grows from 19% at 1 hop to 54% at 4 hops. Claude and the human also agree less at 3 and 4 hops (59% and 53%) than at 1 and 2 hops (73% and 75%), so hard questions are also hard to label. Table XIX shows that the 7B judge detects 27/37 wrong claims at 1–3 hops but only 25/47 at 4–5 hops. A faithfulness score is therefore least reliable on the questions where wrong claims are most frequent.

The silver–human gap. Fig. 5 places both scores side by side. The gap has no single direction: LLM-judge 7B scores higher on human labels, while the verifiers tuned on silver labels fall. Table XX gives per-label results on human labels, using all 165 claims. The human labelled 119 claims supported, 27 unsupported and 19 contradicted, so the contradicted F1 rests on few examples. XLM-R fine-tuned has the best contradicted F1, and the 7B judge the best supported and unsupported F1.

Disagreement in detail. Claude and the human differ on 57 claims. Of these, 29 are claims Claude rejects and the human accepts, 16 go the other way, and 12 differ only between contradicted and unsupported, so both sides call the claim wrong. On the disputed claims the 7B judge sides with the human 28 times and with Claude 20 times. Table XXI shows examples. In several, Claude's reason points to a precise detail,

TABLE XVII
FOUR CLAIMS AND THE VERDICT OF EACH CHECKER (S SUPPORTED, C CONTRADICTED, U UNSUPPORTED; TRANSLATED)

Error type	Claim	Evidence	Silver	7B	3B	Zero-shot	Fine-tuned
Wrong sum	The total fine of the three clinics is 8 million VND.	One clinic alone was fined 30 million VND.	C	S	S	C	C
False denial	The Population Department has no role in inspection or handling violations.	Its duties include specialised inspection and handling violations in the population field.	C	C	S	C	C
Invented item	Fine Vietnam Co. may advertise on Instagram.	Instagram appears only in the certificate of another company in the context.	U	C	U	S	U
Invented date	Circular 30/2017/TT-BYT takes effect on 01/01/2018.	The evidence gives no such date.	U	U	C	U	U

TABLE XVIII
PROFILE BY HOP ON USABLE CLAIMS: SILVER ERROR SHARE, CLAUDE–HUMAN AGREEMENT, MEAN GGF AND CR FOR NAIVE RAG (N) AND GRAPHRAG (G)

Hop	Claims	Wrong	Human	Agree	GGF N	GGF G	CR N	CR G
1	62	19%	44	73%	0.85	0.85	0.04	0.04
2	48	19%	24	75%	0.81	0.84	0.12	0.12
3	52	31%	29	59%	0.83	0.61	0.17	0.21
4	56	54%	34	53%	0.41	0.40	0.33	0.32
5	51	35%	34	68%	0.56	0.46	0.25	0.31

TABLE XIX
DETECTION OF NON-SUPPORTED CLAIMS BY HOP GROUP, ALL USABLE CLAIMS, SILVER LABELS

	1–3 hops		4–5 hops	
Checker	Caught	False al.	Caught	False al.
Judge 7B	27/37	19/125	25/47	4/59
Judge 14B	30/37	20/125	25/48	7/59
Fine-tuned V1	26/37	32/125	22/48	13/59
XLM-R fine-tuned	34/37	54/125	43/48	27/59

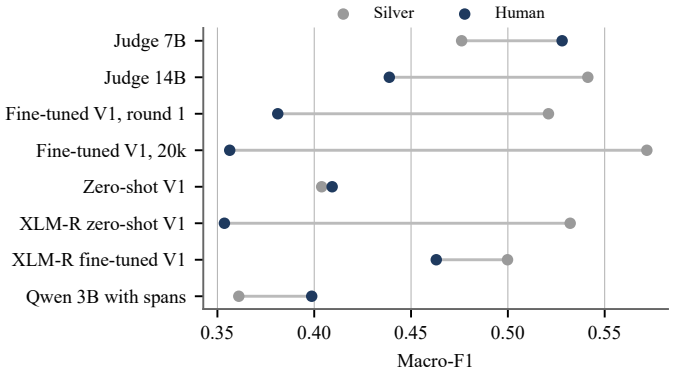


Fig. 5. Macro-F1 on silver labels (test split) and human labels for selected checkers.

TABLE XX
PER-LABEL F1 AND DETECTION ON HUMAN LABELS

Checker	Supp.	Contra.	Unsup.	Caught	False al.
Judge 7B	0.834	0.350	0.400	28/46	21/119
Judge 14B	0.821	0.143	0.353	27/46	23/119
Judge 3B	0.794	0.000	0.255	13/44	20/118
Fine-tuned V1	0.750	0.140	0.254	25/46	35/119
Fine-tuned 20k	0.673	0.158	0.238	27/46	49/119
XLM-R fine-tuned	0.639	0.444	0.306	35/46	58/119
Zero-shot V1	0.747	0.205	0.276	19/46	32/119

such as a shortened name or a change from a subsidy to a contribution, which suggests that the human may accept near matches. In others, Claude asks for an explicit statement where a reader might accept a reasonable inference.

Table XXII shows how two checkers err against the human. The 7B judge tends to accept claims, while XLM-R fine-tuned rejects many supported claims.

Discrimination without thresholds. Fig. 6 shows ROC curves for detecting non-supported claims. On silver labels, fine-tuned verifiers separate from zero-shot, with AUROC 0.738 against 0.545. On human labels the gap narrows but remains (0.650 against 0.568), and XLM-R fine-tuned reaches 0.678. So fine-tuning keeps some ranking signal under human labels even though macro-F1 at the chosen threshold does not improve. At their fixed operating points, both judges lie above the verifier curves on both label sets, so at the same false alarm rate they detect more wrong claims.

Threshold sensitivity. Fig. 7 varies the support threshold. For fine-tuned V1, the dev thresholds give 0.521, while the best grid point on test gives 0.596; for XLM-R fine-tuned the figures are 0.500 and 0.599. Those best points are oracle values, not results, but they show that 51 dev claims select thresholds with an error of up to about 0.1 in macro-F1. Differences of this size between verifiers should not be over-interpreted, and AUROC is the safer comparison.

Tuning thresholds on human labels. If thresholds tuned on silver labels caused the drop, tuning on human labels should recover it. We ran grouped five-fold cross-validation over the 165 human-labelled claims, keeping claims of one question in one fold, and pooled the held-out predictions (Table XXIII). Fine-tuned V1 does not improve (0.381 to 0.360, CI of the difference $[-0.08, 0.04]$), and no verifier improves significantly. Thresholds alone therefore do not explain the gap on this sample, although each fold trains on only about 130 claims and the chosen thresholds vary between folds.

Agreement between checkers. Fig. 8 gives pairwise Cohen's κ on the human-labelled claims. The human agrees more with the 7B judge ($\kappa = 0.34$) than with the silver labels ($\kappa = 0.29$) or the 14B judge ($\kappa = 0.26$). Agreement with the human is low for every checker, and the 7B judge and the

TABLE XXI
EXAMPLES OF DISAGREEMENT BETWEEN CLAUDE AND THE HUMAN (TRANSLATED; S SUPPORTED, C CONTRADICTED, U UNSUPPORTED)

Claim	Claude's reason	Claude	Human	7B
The representative of Thai Ha General Clinic is Mr. Le Son.	The correct name is Mr. Le Quang Son.	C	S	S
Policy sets the insurance contribution of students at 50% of the minimum contribution.	Students receive a subsidy of at least 50%; they do not contribute 50%.	C	S	S
The Institute of Strategy and Policy manages the Hai Thuong Lan Ong award golden book.	The golden book is managed by the Department of Traditional Medicine.	C	S	C
Fine Vietnam Co. may advertise on Pinterest.	Pinterest is not mentioned.	U	C	C
Health staff at the CDC centre must propose the GSP certification of vaccine stores.	No rule says that health staff propose it.	U	S	S
GSP principles are taken from the World Health Organization.	Issued on the basis of WHO guidance.	S	C	S

TABLE XXII
HUMAN LABELS (ROWS) AGAINST PREDICTIONS (COLUMNS) ON THE HUMAN-LABELLED CLAIMS

Judge 7B	Supp.	Contra.	Unsup.
supported	98	9	12
contradicted	7	7	5
unsupported	11	5	11
XLM-R fine-tuned	**Supp.**	**Contra.**	**Unsup.**
supported	61	16	42
contradicted	4	12	3
unsupported	7	7	13

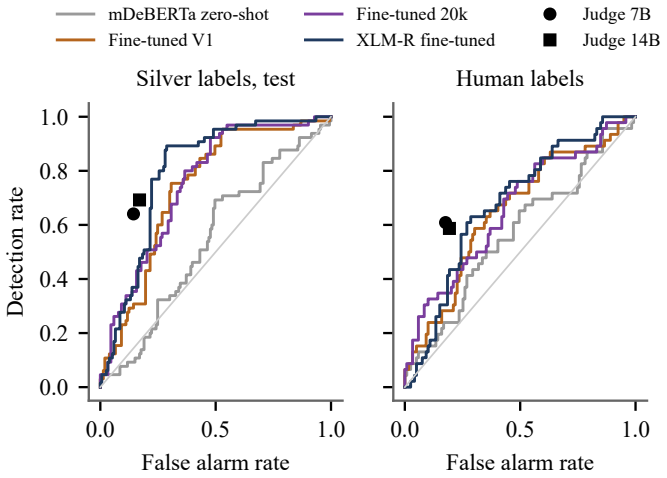


Fig. 6. ROC curves for detecting non-supported claims. Dots mark the fixed operating points of the 7B and 14B judges.

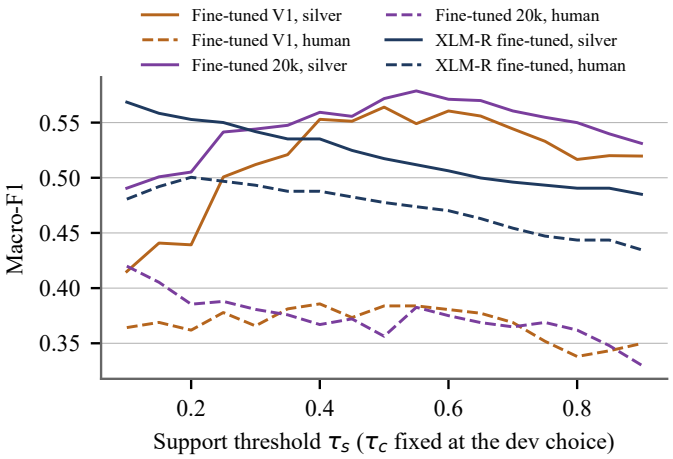


Fig. 7. Macro-F1 against the support threshold. Solid: silver labels on test; dashed: human labels.

TABLE XXIII
MACRO-F1 ON HUMAN LABELS: THRESHOLDS FROM SILVER DEV AGAINST GROUPED FIVE-FOLD THRESHOLDS FROM HUMAN LABELS

Verifier	Silver	Human	Diff.	95% CI
mDeBERTa zero-shot	0.409	0.332	−0.077	[−0.14, −0.02]
Fine-tuned V1	0.381	0.360	−0.022	[−0.08, 0.04]
Fine-tuned 20k	0.356	0.373	0.017	[−0.04, 0.07]
XLM-R zero-shot	0.354	0.354	0.001	[−0.01, 0.02]
XLM-R fine-tuned	0.463	0.445	−0.018	[−0.09, 0.05]

verifiers also agree little with one another, which suggests that they make different errors.

Combining two checkers. Because the 7B judge and XLM-R fine-tuned disagree often ($\kappa = 0.24$), we tested two simple rules on the binary task of flagging a claim (Table XXIV). Flagging when either checker objects catches 61/64 wrong claims on silver labels at the cost of 72/153 false alarms. Flagging only when both object cuts false alarms to 14/153 and still catches 37. Neither rule beats the 7B judge alone on flag F1, but they offer two useful operating points.

Answer-level agreement. Fig. 9 plots RAGAS against reference GGF. Scores spread widely at every GGF level, so even the best correlation does not make a single answer's score dependable. Two RAGAS versions also differ: their rank correlation with each other is $\rho = 0.54$ on 61 answers.

Results by system. Table XXV splits checker scores by the system that produced the answer. Differences between systems are smaller than differences between label sets, so the checker comparison does not depend on one system.

Sample size. Many intervals in this paper include zero. Table XXVI estimates the claims needed to detect a macro-F1 difference of 0.05 with 80% power, taking the standard error from the bootstrap interval and assuming that it shrinks with the square root of the sample. The estimates are optimistic because claims from one answer are not independent. They

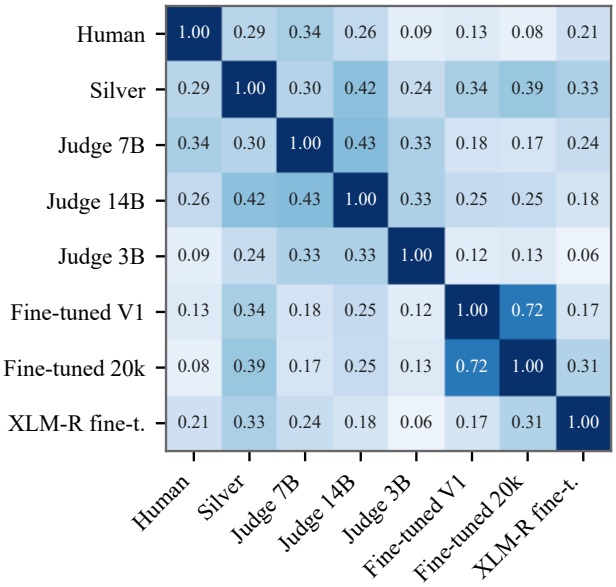


Fig. 8. Pairwise Cohen's κ on the human-labelled claims.

TABLE XXIV
FLAGGING RULES FROM THE 7B JUDGE AND XLM-R FINE-TUNED V1; "F1" IS FOR THE FLAG CLASS

	Silver, test			Human		
Rule	Caught	False al.	F1	Caught	False al.	F1
7B only	41/64	22/153	0.646	28/46	21/119	0.589
XLM-R only	57/64	64/153	0.616	35/46	58/119	0.504
either flags	61/64	72/153	0.619	39/46	63/119	0.527
both flag	37/64	14/153	0.643	24/46	16/119	0.558

still show that the target of about 3,000 claims in our proposal is of the right order, and that the present sample cannot settle differences of this size.

Two-tier checking. A cheap verifier can decide the claims it is sure about and pass only the uncertain ones to the 7B judge. We rank claims by the distance between the verifier's support probability and its threshold, and send the closest share to the judge (Table XXVII). With fine-tuned V1 and half of the claims sent to the judge, the combination reaches 0.530 on human labels, level with the judge alone (0.528), while false alarms on silver labels fall from 22/153 to 13/153. The budget of one half was chosen after seeing these curves and should be checked on new data, but the result suggests that the expensive judge is not needed for every claim.

Round 4: a larger judge, more systems, another domain and a Verify operator. All numbers in this paragraph use silver labels unless stated. A 32B judge, run on Kaggle, reaches 0.571 on silver labels but only 0.377 on human labels, significantly below the 7B judge (difference −0.151, CI [−0.26, −0.03]): a larger judge moves closer to the LLM labeller and further from the human, the fourth instance of the pattern above. Three more systems (BM25, dense retrieval, and Naive RAG with Qwen2.5-7B) answered the same 40 questions and added 413 silver-labelled claims. Silver GGF

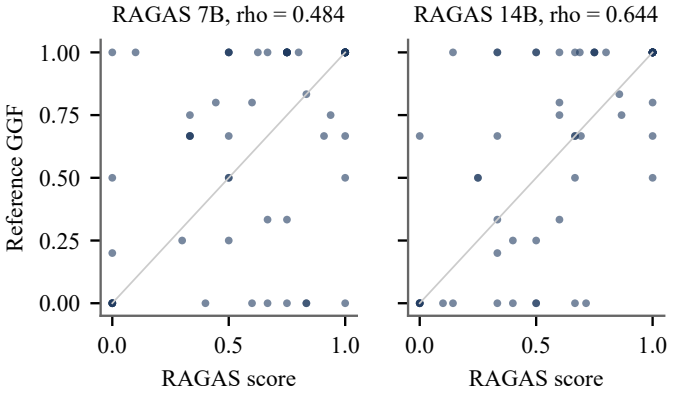


Fig. 9. RAGAS score against reference GGF for test answers. Each dot is one answer; the grey line marks equality.

TABLE XXV
MACRO-F1 BY ANSWER SYSTEM

	Silver, test		Human	
Checker	Naive	Graph	Naive	Graph
Judge 7B	0.442	0.498	0.532	0.515
Judge 14B	0.536	0.541	0.477	0.389
Fine-tuned V1	0.515	0.523	0.408	0.357
Fine-tuned 20k	0.533	0.596	0.385	0.330
XLM-R fine-tuned	0.477	0.519	0.482	0.448

ranks the five systems Naive 0.694, GraphRAG 0.638, Naive-7B 0.616, BM25 0.575, dense 0.532, while GGF from the 7B judge ranks them almost in reverse (Spearman −0.60 over five systems) and scores dense retrieval highest. On the new claims, XLM-R of round 3 reaches 0.563 and the 7B judge 0.525. On a 30-question pilot over higher-education quality-assurance guidelines (ViEduReg, 77 claims), the 7B judge keeps 0.527 while XLM-R trained on health texts falls to 0.269, below the always-supported baseline, because it flags 34/55 correct claims. Finally, a KAG-style Verify operator checks each claim with XLM-R, sends uncertain claims to the 7B judge (27 calls for 284 claims) and checks cited document numbers, then revises the answer once or refuses. As a checker it scores 0.479 on human labels, not significantly different from the 7B judge. Rescored by an independent 14B judge, GGF rises from 0.691 to 0.805, level with the earlier revise loop (0.806), but the operator refuses 31 of 80 answers, 13 of which were fully supported.

Effect of the audit pass. Table XXVIII compares pass C, stratified with full passages, against the random quick passes. The 7B judge leads in both, so the main finding does not depend on the display condition. Pass C oversamples errors, so scores between the two columns are not directly comparable.

Cost. Table XXIX lists wall-clock times of round 2. They come from file timestamps of a single run, so they are indicative only. The 14B judge with RAGAS took about 2.8 hours for 80 answers, roughly 30 times the 3B span run, without a significant claim-level gain.

TABLE XXVI
ESTIMATED CLAIMS NEEDED TO DETECT A DIFFERENCE OF 0.05 IN MACRO-F1 WITH 80% POWER

Comparison	n	SE	Needed
Fine-tuned V1 − judge 7B, silver	217	0.057	2,200
Judge 14B − judge 7B, silver	217	0.058	2,300
Judge 14B − judge 7B, human	165	0.058	1,800
XLM-R fine-tuned − judge 7B, human	165	0.055	1,500

TABLE XXVII
TWO-TIER CHECKING: SHARE OF CLAIMS SENT TO THE 7B JUDGE AGAINST MACRO-F1 AND FALSE ALARMS

	Fine-tuned V1 first			XLM-R fine-tuned first		
To judge	Silver	False al.	Human	Silver	False al.	Human
0%	0.524	37/153	0.381	0.496	64/153	0.463
30%	0.500	22/153	0.494	0.517	59/153	0.506
50%	0.491	13/153	0.530	0.511	61/153	0.503
70%	0.487	18/153	0.523	0.484	43/153	0.507
100%	0.476	22/153	0.528	0.476	22/153	0.528

TABLE XXVIII
MACRO-F1 ON HUMAN LABELS BY AUDIT PASS

Checker	Pass C (60)	G–J (105)	All (165)
Judge 7B	0.525	0.528	0.528
Fine-tuned V1	0.452	0.331	0.381
Zero-shot V1	0.510	0.330	0.409
Judge 3B	0.380	0.328	0.349

TABLE XXIX
WALL-CLOCK TIME OF ROUND 2 JOBS (SINGLE RUN)

Job	GPU	Time
Judge and RAGAS 14B, 80 answers	T4	about 2.8 h
Qwen 3B with spans, 80 answers	T4	5 min
XLM-R zero-shot, 284 claims, V1 and V3	T4	6.5 min
XLM-R fine-tune, 16,000 pairs, with scoring	T4	69 min
mDeBERTa fine-tune, 29,993 pairs, 900 chars, with scoring	T4	53 min
mDeBERTa fine-tune, 29,993 pairs, 450 chars	GTX 1650	about 2 h

VII. DISCUSSION

Which checker should a practitioner use? On this data, the 7B judge is the most defensible default. It leads on human labels, is not significantly below the best verifier on silver labels, and needs no training. Moving to 14B brings a clear gain only at answer level, where RAGAS reaches $\rho = 0.644$, and even that gain is untested. A team that wants one faithfulness number per answer can use RAGAS 14B, while a team that needs to locate wrong claims should stay with a claim-level judge. For regulations, where missing a wrong claim costs more than reviewing a correct one, the either-flags rule is a reasonable triage step: it passes few wrong claims but sends close to half of the correct claims to a human reader. It supports human review and does not replace it.

Why do silver and human labels disagree? We can see the size of the disagreement but not who is right. Claude labels 59 claims as non-supported and the human confirms 18 of them. Three explanations are plausible. Claude may be stricter about implicit inferences. The annotator, who is not a legal expert, may accept claims that a lawyer would reject. The quick passes may also hide distant evidence, although Table XXVIII shows the same leading checker in both conditions. Adjudication by two legal experts is the only way to separate these. Until then, both label sets should be reported, because a ranking from one source does not transfer to the other.

What does fine-tuning learn? Accuracy on synthetic dev pairs reaches about 0.9, far above any score on real claims. Synthetic errors change a number, a date or a negation, or borrow a sentence from elsewhere. Real errors in our sample also include wrong subjects, wrong aggregation across documents and invented list items, which these perturbations do not produce. A verifier can thus master the synthetic task without learning the real one. This fits the flat AUROC from 15,000 to 30,000 pairs: adding more pairs of the same kind does not help. Round 3 reproduced the observed error types; the verifier learned them on held-out pairs, yet its human-label score did not rise.

Why did graph input not help? Prepending metadata to each window makes every claim that names a document look entailed. Selective triples avoid part of this, but the released graph is coarse and is passed as plain text. Our result concerns this form of graph input on this graph, and it does not show that graphs cannot help verification.

What does revision really do? A faithfulness score alone rewards deletion. An answer that says less contains fewer wrong claims, so GGF improves as content disappears. Any revision loop should therefore be scored by faithfulness and completeness together, with an independent scorer. With a 3B generator the loop mostly removes content, and the Verify operator of round 4 mostly refuses; a larger generator may behave differently.

Recommendations for meta-evaluation. Our results suggest four practices for future work on Vietnamese faithfulness evaluation. First, report the label source and keep human labels for at least part of the sample, because a ranking on LLM labels may not hold. Second, compare checkers with threshold-free measures as well as macro-F1, since small dev sets make thresholds unstable. Third, report detection and false alarms separately, because a high macro-F1 can hide a verifier that rejects a third or more of correct claims. Fourth, split results by hop, since a mean score hides the weakest questions, where errors are most common.

VIII. THREATS TO VALIDITY

Silver reference. Claude agrees with the human on 65% of claims ($\kappa = 0.29$). We therefore report both label sets; more human labels and a second annotator with adjudication are needed.

One annotator. The annotator is not a legal expert, and quick passes show 4 evidence sentences, so distant evidence may be missed. Pass C uses full passages and is reported separately.

Small scale. We evaluate five systems, only two of them with human labels, 697 claims and judges up to 32B. Bootstrap intervals and paired tests accompany the main comparisons, without multiple-comparison correction.

Thresholds tuned on silver labels. Only 51 dev claims are available, and thresholds tuned on silver labels may lean toward Claude. Tuning on human labels did not improve scores in our cross-validation (Table XXIII), but that sample is small.

Revision scored by LLMs. Round 1 revisions are scored by the judge that flagged them; round 4 rescores with an independent 14B judge, which is still an LLM. Human scoring is needed.

Graph as text. Triples are appended as text, so verifiers do not read graph structure. ViEduReg is a 30-question pilot with silver labels, and the three systems of round 4 have silver labels only, written on short evidence excerpts.

Post hoc and multiple comparisons. Many comparisons were run without correction, and the analyses in Section VI-F were chosen after seeing the main results. Intervals that barely exclude zero should be read as hints.

Implementation. Early fine-tuning runs produced NaN loss because the mDeBERTa checkpoint is stored in float16; all reported runs use float32. Round 1 on Colab used transformers 5.x, and round 2 used 4.46. Ollama runs Q4_K_M quantised models, so results may differ from full-precision weights.

IX. Conclusion

On Vietnamese healthcare regulations, automatic faithfulness metrics running on small open LLMs match reference labels only at a low to moderate level. The 7B judge is the most stable metric across both label sets; the 14B judge is not significantly better at claim level, and the 32B judge is significantly worse on human labels. RAGAS is best at answer level and improves with model size. Fine-tuning mDeBERTa on synthetic in-domain data raises silver scores but not human-label scores, and about 20,000 pairs bring no significant gain; XLM-R fine-tuning improves on human labels but raises many false alarms. Tuning thresholds on human labels did not close the gap, and neither did hard negatives shaped like the observed errors, which gave the best silver score of the study. A two-tier checker that sends only uncertain claims to the judge kept the judge's human-label score at half the cost. Meta-evaluation with LLM labels can therefore change metric rankings, and reports should state their label source. Revising answers, including with a KAG-style Verify operator, reduces wrong claims by deleting content or refusing to answer, which is a trade-off and not a pure improvement.

Future work includes full human labelling by two annotators with legal expertise and threshold selection on human labels; a fine-tuned small Qwen verifier whose spans are copied from the evidence; structured triple input for verifiers and extension to LightRAG and HippoRAG 2; human labels for ViEduReg and the round-4 systems; and human scoring of revised answers. Code and data are available at https://github.com/vinhha195-oss/vikag-faith.

Appendix A
Prompts

The RAGAS prompts follow the library. The statement prompt reads: "Given a question, an answer, and sentences from the answer analyze the complexity of each sentence given under 'sentences' and break down each sentence into one or more fully understandable statements while also ensuring no pronouns are used in each statement." The verdict prompt reads: "Your task is to judge the faithfulness of a series of statements based on a given context. For each statement you must return verdict as 1 if the statement can be directly inferred based on the context or 0 if the statement can not be directly inferred based on the context." Both ask for JSON output, and the first keeps the language of the answer.

The remaining prompts are in Vietnamese; these are English translations.

Generator. "You are an assistant answering questions about Vietnamese healthcare legal documents. Use only information in the CONTEXT. Answer in Vietnamese in 2–5 sentences, naming documents when available. If the context is insufficient, say clearly that no information was found."

Claim extraction. "Split the answer into atomic claims. Each claim states ONE fact with one main predicate. Each claim must be self-contained: replace pronouns with specific names. Keep numbers, dates and document identifiers as in the answer. Add nothing beyond the answer, and drop sentences that only say no information was found." Two worked examples follow.

LLM-judge. "You are a fact checker. For each claim, compare with the EVIDENCE and assign exactly one label. supported: the evidence implies the claim. contradicted: the evidence says the opposite (wrong figure, date, document name, negation, and so on). unsupported: the evidence does not mention it or is not enough to conclude." The span verifier adds: "For each claim, copy verbatim one short sentence from the EVIDENCE as the basis in the span field; if there is none, leave the string empty."

Appendix B
Annotation Protocol

The annotator received a single offline HTML page holding the claim, the question and the evidence; labels were saved locally and exported as a file. Claim identifiers were anonymised, so the annotator could not tell which system produced a claim, and neither Claude's labels nor its reasons were shown. An earlier pre-filled review pass confirmed all 284 labels with 0 changes in about 10 minutes. Such a result says little about label quality, so we discarded it and moved to blind passes. Passes D and E were excluded because their exports were not created through the labelling page. In the quick passes, the annotator could skip a claim; 15 claims were skipped and are not counted.

APPENDIX C
FULL RESULTS

Table XXX lists every round 1 configuration. Table XXXII lists answer-level correlations, Table XXXIII the dev thresholds, and Table XXXI the paired comparisons of round 2.

APPENDIX D
REPRODUCIBILITY

The repository holds the pipeline as numbered scripts. `s1_answer` generates answers, `s2_claims` extracts claims, `s3_pool` builds the evidence sets, `s3b_triples` builds triples and the mutual index, `s4_judges` runs RAGAS and the LLM-judges, `s5_verifier` prepares data and trains and scores verifiers, `s6_annotation` and `s6_quick` build the labelling pages, `s7_analysis` computes the round 1 results, `s8_revise` runs verify-then-revise, `s9_selfcheck` runs SelfCheckGPT, `s10_dot2` analyses round 2, `s11_ieee_extra` and `s12_ieee_more` compute the post hoc analyses of this paper, `s13_cascade` to `s18_dot3` cover round 3, and `s19_dot4` to `s22_record` cover round 4. All sampling uses seed 42, and every number in this paper is generated from the stored result files.

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TABLE XXX
ALL ROUND 1 CONFIGURATIONS ON 218 TEST CLAIMS (SILVER LABELS), WITH MACRO-F1 ON HUMAN LABELS

Method	Macro-F1	95% CI	Acc.	AUROC	F1 supp.	F1 contra.	F1 unsup.	Human
LLM-judge 7B	0.476	[0.40, 0.55]	0.687	0.748	0.853	0.321	0.254	0.528
LLM-judge 3B	0.439	[0.37, 0.53]	0.692	0.643	0.825	0.129	0.364	0.349
Fine-tuned V1	0.521	[0.44, 0.60]	0.656	0.738	0.773	0.379	0.410	0.381
Fine-tuned V1 + rule	0.502	[0.42, 0.58]	0.638	0.631	0.761	0.355	0.390	0.382
Fine-tuned V2	0.465	[0.39, 0.54]	0.633	0.735	0.784	0.226	0.385	0.336
Fine-tuned V2 + rule	0.457	[0.38, 0.53]	0.619	0.662	0.771	0.215	0.385	0.338
ViNLI only V1	0.405	[0.33, 0.47]	0.624	0.649	0.766	0.345	0.105	0.395
ViNLI only V1 + rule	0.396	[0.33, 0.46]	0.610	0.546	0.755	0.328	0.105	0.391
ViNLI only V2	0.332	[0.28, 0.38]	0.619	0.644	0.766	0.230	0.000	0.346
ViNLI only V2 + rule	0.326	[0.28, 0.37]	0.606	0.516	0.760	0.219	0.000	0.355
Zero-shot V1	0.404	[0.33, 0.47]	0.592	0.545	0.737	0.308	0.167	0.409
Zero-shot V1 + rule	0.401	[0.33, 0.47]	0.587	0.534	0.733	0.302	0.167	0.413
Zero-shot V2	0.329	[0.27, 0.39]	0.693	0.534	0.815	0.171	0.000	0.295
Zero-shot V2 + rule	0.322	[0.27, 0.38]	0.679	0.520	0.812	0.154	0.000	0.314
SelfCheckGPT-NLI	0.409	[0.34, 0.48]	0.463	0.600	0.565	0.400	0.262	0.336
Rule only	0.274	[0.26, 0.29]	0.688	0.513	0.822	0.000	0.000	0.310
Always supported	0.275	[0.26, 0.29]	0.702	0.500	0.825	0.000	0.000	0.279

TABLE XXXI
PAIRED MACRO-F1 DIFFERENCES OF ROUND 2 METHODS WITH 95% CI (1,000 BOOTSTRAP RESAMPLES, NOT CORRECTED FOR MULTIPLE COMPARISONS)

Method	Silver, − fine-tuned V1	Silver, − judge 7B	Human, − fine-tuned V1	Human, − judge 7B
Judge 14B	0.020 [−0.07, 0.11]	0.069 [−0.05, 0.18]	0.058 [−0.03, 0.14]	−0.089 [−0.20, 0.03]
Zero-shot V3	−0.098 [−0.19, −0.01]	−0.051 [−0.14, 0.04]	−0.053 [−0.13, 0.03]	−0.200 [−0.30, −0.09]
20k, 450 chars V1	−0.080 [−0.16, 0.00]	−0.033 [−0.15, 0.08]	−0.011 [−0.08, 0.06]	−0.158 [−0.27, −0.05]
20k, 450 chars V3	−0.020 [−0.08, 0.04]	0.027 [−0.09, 0.14]	0.022 [−0.04, 0.08]	−0.124 [−0.25, 0.00]
20k, 900 chars V1	0.051 [−0.02, 0.12]	0.098 [−0.02, 0.22]	−0.025 [−0.06, 0.01]	−0.172 [−0.28, −0.05]
20k, 900 chars V3	0.038 [−0.05, 0.12]	0.085 [−0.03, 0.19]	−0.035 [−0.08, 0.01]	−0.182 [−0.30, −0.07]
XLM-R zero-shot V1	0.011 [−0.08, 0.11]	0.052 [−0.05, 0.15]	−0.028 [−0.10, 0.05]	−0.174 [−0.29, −0.07]
XLM-R zero-shot V3	−0.005 [−0.10, 0.10]	0.036 [−0.07, 0.14]	−0.012 [−0.08, 0.06]	−0.158 [−0.27, −0.05]
XLM-R fine-tuned V1	−0.021 [−0.12, 0.08]	0.020 [−0.07, 0.11]	0.082 [−0.02, 0.18]	−0.065 [−0.17, 0.04]
XLM-R fine-tuned V3	−0.044 [−0.14, 0.05]	−0.003 [−0.09, 0.09]	0.052 [−0.05, 0.15]	−0.095 [−0.19, 0.01]
Qwen 3B with spans	−0.163 [−0.26, −0.06]	−0.129 [−0.22, −0.04]	0.017 [−0.10, 0.13]	−0.133 [−0.22, −0.06]

TABLE XXXII
ANSWER-LEVEL CORRELATION WITH REFERENCE GGF ON THE TEST SPLIT, SORTED BY SPEARMAN ρ; n IS THE NUMBER OF ANSWERS WITH VALID OUTPUT

Metric	Spearman	Kendall	n
RAGAS 14B	0.644	0.527	61
LLM-judge 14B	0.597	0.533	63
RAGAS 7B	0.484	0.396	62
LLM-judge 7B	0.418	0.387	63
Fine-tuned V2	0.383	0.318	63
LLM-judge 3B	0.367	0.322	61
SelfCheckGPT-NLI	0.270	0.230	63
RAGAS 3B	0.258	0.214	62
Zero-shot V2	0.161	0.145	63
Rule only	0.147	0.132	63
Fine-tuned V1	0.142	0.121	63
Zero-shot V1	0.003	0.007	63
ViNLI only V1	−0.005	0.000	63
ViNLI only V2	−0.029	−0.019	63

TABLE XXXIII
THRESHOLDS CHOSEN ON THE DEV SPLIT

Verifier	τ_s	τ_c
Zero-shot V1	0.65	0.45
Zero-shot V2	0.15	0.10
Zero-shot V3	0.75	0.15
Fine-tuned V1	0.35	0.10
Fine-tuned V2	0.30	0.10
ViNLI only V1	0.35	0.25
ViNLI only V2	0.20	0.10
SelfCheckGPT	0.80	0.15
20k, 450 chars V1	0.10	0.10
20k, 450 chars V3	0.15	0.30
20k, 900 chars V1	0.50	0.40
20k, 900 chars V3	0.55	0.40
XLM-R zero-shot V1	0.65	0.10
XLM-R zero-shot V3	0.80	0.65
XLM-R fine-tuned V1	0.65	0.10
XLM-R fine-tuned V3	0.65	0.10